

1. The RPG Hero

You have a player dictionary: `player = {"name": "Aragon", "level": 10, "health": 50}`.

- **Task:** If the player drinks a "Healing Potion," their health should increase by 20. Update the dictionary and print the new health.

2. Factory Machine Monitoring

A factory monitors machines during production.

Task:

- Check machine status continuously
- If status = "Maintenance" → skip
- If status = "Failure" → stop production
- Count machines running normally

3. The Smart Elevator

An elevator can only carry 500kg. You are adding luggage weights one by one from a list:

`luggage = [100, 200, 50, 150, 100]`.

- **Task:** Use a loop to add weights to a `current_weight` variable. If the next bag will put the total over 500kg, **break** the loop and print "Elevator Full."

4. Bank Loan Eligibility Checker

A bank reviews loan applications.

Task:

For each applicant:

- If income < minimum → reject
- If credit score very low → stop approvals for review
- Otherwise approve
- Count approved loans

5. Result Processing

A teacher enters marks for students.

Task:

- Input marks for N students
- Assign grades:
 - $\geq 80 \rightarrow A$
 - $\geq 60 \rightarrow B$
 - $\geq 40 \rightarrow C$
 - $< 40 \rightarrow \text{Fail}$
- Count how many students failed

6. The Contact Book

You have a dictionary of contacts: `contacts = {"Alice": "555-1234", "Bob": "555-5678"}`.

- **Task:** Check if "Charlie" exists in the dictionary. If not, add him with the number "555-0000".



Exam Instruction

1. This exam contains **6 questions** in total.
2. You must answer **any 5 questions**.
3. Each question carries equally **10** marks.
4. Read all questions carefully before answering.
5. Write clean, well-structured, and properly indented code.
6. Use meaningful variable names where applicable

Submission Instructions (GitHub)

1. Create a **public GitHub repository** for this exam.
 2. Create separate files for each answer: `q1.py/q2.py`
 - Only upload the **5 questions you answered**.
 3. Push your code before the time ends (60 minutes).
 4. After submission, share the **GitHub repository link** in the exam submission form / classroom portal.
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